

MODULE 8

The shell and signals

tms-client · app.config · signal · computed · @if

TMS client

```
@Component({ standalone: true })
```

```
earnedCredits = signal(45);
```

```
<p>{{ earnedCredits() }}</p> // () = read  
the value
```

Give the API a browser face

Session goals

By the end of this session, you will:

- Contrast .NET (server JSON) with Angular (browser UI) for the TMS
- Scaffold tms-client with the CLI flags and explain each flag
- Read app.config.ts as global wiring change detection, router + input binding, HttpClient
- Explain why signal() + () in templates drives repaints; demo plain field vs signal
- Add computed() and @if blocks for Liya-style dashboard copy
- Spot first-hour traps: missing (), NullInjector, missing router-outlet

The problem

You built a backend API that works perfectly. Postman proves it. Integration tests confirm it. Scalar documents every endpoint.

But nobody uses Postman to check their enrollment status.

Students open a browser. Instructors open a browser. Admins open a browser.

Right now, the browser shows raw JSON. The TMS has no face.

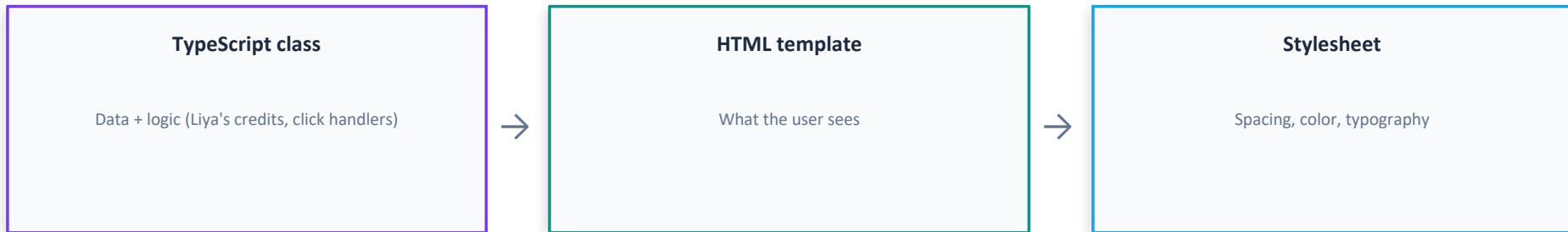
What is Angular?

	.NET (Backend)	Angular (Frontend)
Where it runs	On the server	In the user's browser
Language	C#	TypeScript
What it produces	JSON responses	A visual, interactive page
Who uses it	Postman, other apps, the frontend	The actual human being

SPA in one sentence

One HTML load; Angular swaps views and handles clicks without full page reloads. Liya's dashboard to course detail feels instant.

What is a component?



- Fundamental building block same idea as a C# class, but with a visible template
- Pages compose many components (header, course cards, sidebar) fix the card in isolation

What is the Angular CLI?

.NET	Angular
dotnet new webapi	ng new tms-client
dotnet run	ng serve
dotnet add package	npm install
Controllers/ + Program.cs	src/app/ + app.config.ts

ng

The CLI scaffolds files, compiles TypeScript, and runs the dev server you fill in TMS behavior, not boilerplate from scratch.

Angular 22: what changed

- If you Google "Angular tutorial," half the results show `app.module.ts` that file no longer exists
- Standalone components each component declares its own dependencies; no central `NgModule` registry
- Zoneless change detection Angular does not spy on every field; repaints follow explicit signal reads
- Vitest default test runner; fast Node runs instead of slow browser harnesses

The application shell `app.config.ts`

app.config.ts Program.cs of the SPA

```
import { ApplicationConfig, provideZoneChangeDetection } from "@angular/core";
import { provideRouter, withComponentInputBinding } from "@angular/router";
import { provideHttpClient } from "@angular/common/http";
import { routes } from "../app.routes";

export const appConfig: ApplicationConfig = {
  providers: [
    provideZoneChangeDetection({ eventCoalescing: true }),
    provideRouter(routes, withComponentInputBinding()),
    provideHttpClient()
  ]
};
```

- 1) Change detection when the DOM is allowed to update (batch rapid UI events with coalescing)
- 2) Routing which component owns /dashboard, /courses/:id, ...
- 3) HttpClient calling the .NET TMS API (wired now; used heavily in Lab 6)

Scaffold and run tms-client

Commands (from spine)

```
ng new tms-client --style=scss --ssr=false --standalone --routing  
  
cd tms-client  
  
ng serve
```

Flag	What it does
--style=scss	SCSS instead of plain CSS
--ssr=false	No SSR TMS is login-protected; skip indexing complexity
--standalone	No legacy app.module.ts Angular 22 default shape
--routing	app.routes.ts + router-outlet from day one

After ng serve

Strip the welcome template, keep <router-outlet /> in app shell blank canvas before Labs 2–4 navigation work.

Signals variables the screen watches

Plain field (in a zoneless app spine)

- `let earnedCredits = 45;`
- `addCredits() { earnedCredits += 3; }`
- Spine: memory updates; screen may not (zoneless)
- Signals lesson still applies on every stack

signal() (working path)

- `earnedCredits = signal(45);`
- `addCredits() { this.earnedCredits.update(c => c + 3); }`
- Same click DOM updates
- Angular traces reads in the template

Reading signals in templates

Template (spine)

```
<!-- CORRECT  calls the signal, gets 45 -->
```

```
<p>Credits: {{ earnedCredits() }}</p>
```

```
<!-- WRONG  renders the signal wrapper object -->
```

```
<p>Credits: {{ earnedCredits }}</p>
```

- A signal is a function that holds a value call it with () to read
- Half the room will print [object Object] once fix once, they fix it everywhere (input, computed, model later)

Derived values with computed()

student-dashboard.ts (pattern)

```
earnedCredits = signal(45);

graduationStatus = computed(() =>

  this.earnedCredits() >= 120 ? 'Eligible' : 'In Progress'

);
```

- When earnedCredits moves 45 → 48, graduationStatus recalculates no manual refresh
- Spreadsheet mental model cell B depends on cell A; change A, B recomputes
- No lifecycle hook, no event bus computed caches until sources move

Modern control flow @if

Template (spine no *ngIf import)

```
@if (earnedCredits() >= 120) {  
  <div class="success">You are eligible to graduate.</div>  
} @else if (earnedCredits() >= 90) {  
  <div class="warning">Final year  consult your advisor.</div>  
} @else {  
  <div class="info">Keep working. {{ 120 - earnedCredits() }} credits remaining.</div>  
}
```

- Cleaner to read built into the compiler; no *ngIf import
- @else if is first-class; smaller compiled output than legacy structural directives

Common mistakes Session 1

Mistake	What you see	How to fix
Missing () on signals	[object Object] or blank display	Add (): {{ earnedCredits() }}
Missing providers in app.config.ts	NullInjectorError: No provider for Router	Add the missing provideX() function
Using app.module.ts from old tutorials	NgModule decorator not found errors	Delete the file. This project is standalone.
Missing <router-outlet />	Routes configured but page is blank	Add the outlet in app.html

Session 1 summary

- Angular is the TMS face in the browser SPA, components, CLI
- app.config.ts registers change detection, router (+ input binding), HttpClient
- signal / computed / @if are the reactive spine for Liya's dashboard
- Parentheses on reads the universal first-session fix

Next session preview

Session 2 splits the dashboard into CourseCardComponent input(), output(), imports[], @for + track, lazy routes.

Next: Session 2 components and routing

input() / output(), parent imports, @for + track course.id, loadComponent routes, router-outlet as the paint canvas.